

Sikun Xu

CONTACT INFORMATION

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Olin Business School
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RESEARCH INTERESTS

Machine Learning and AI Systems; Causal Inference; Data-Driven Decision-Making; Recommendation Systems; LLM and Generative Models

EDUCATION

Washington University in St. Louis

- Ph.D. Candidate, Olin Business School 2021–Present
- Advisors: Raphael Thomadsen and Dennis Zhang

Columbia University in the City of New York

- M.S. in Operations Research 2019–2020

Shanghai Jiao Tong University

- B.S. in Industrial Engineering 2015–2019

JOB MARKET PAPERS

- [The Winner’s Curse in Data-Driven Decision Making: Evidence and Solutions](#), with Raphael Thomadsen and Dennis Zhang. *Submitted*.
- SOTA or Luck? The Winner’s Curse in LLM Leaderboards, with Heng Zhang, Raphael Thomadsen, and Dennis Zhang. *Submitted*.

WORKING PAPERS

1. [A Causal Approach to Representation Learning for Unstructured Data](#), with Zhenling Jiang, Zhengling Qi, and Dennis Zhang. *Major revision at **Management Science***.

CONFERENCE PROCEEDINGS

1. Xu, Sikun, Ruoyi Ma, Daniel K. Molzahn, Hassan Hijazi, and Cédric Jozs. “[Verifying global optimality of candidate solutions to polynomial optimization problems using a determinant relaxation hierarchy](#).” In 2021 60th IEEE Conference on Decision and Control (CDC), pp. 3143-3148. IEEE, 2021.

WORK IN PROGRESS

1. From Prompts to Policies: Offline Policy Learning with Noisy Generative Implementation
2. Quality-Gated Interest Exploration in Online Personalized Recommendation, with industry partners
3. Generative Learning-to-Rank in Recommendation Systems, with industry partners

INDUSTRY EXPERIENCE

- **TikTok Inc.**, San Jose, Machine Learning Scientist Intern 2026–Present
- **Ask2.AI**, New York, Co-founder and Consultant 2022–Present
- **Sauma Capital LLC**, New York, Quantitative Researcher 2021
- **Capstone Investment Advisors**, New York, Investment Intern 2020
- **Cardinal Operations**, Shanghai, Algorithm Engineer Intern 2018

TEACHING
EXPERIENCE

Guest Lecturer

- MGT 680E AI & Machine Learning Business Applications, Washington University in St. Louis Fall 2024
- IEOR 4721 AI Application in Finance, Columbia University Spring 2022
- IEOR 4721 AI Application in Finance, Columbia University Summer 2021

Teaching Assistant

- Olin Business School, Washington University in St. Louis
 - MGT 680E AI & Machine Learning Business Applications
 - SCOT 519E Revenue Management
 - SCOT 500M Supply Chain Analytics: Stochastic Models
 - SCOT 558 Advanced Operations Strategy
 - SCOT 5704 Operations Management
 - SCOT 356 Operations and Manufacturing Management
 - SCOT 400D Supply Chain Management
 - SCOT 500D Project Management
- Columbia University in the City of New York
 - IEOR 4742 Deep Learning
 - IEOR 4525 Machine Learning

CONFERENCE
PRESENTATIONS

The Winner's Curse in Data-Driven Decision Making: Evidence and Solutions

- INFORMS Annual Meeting, San Francisco, CA (Expected) 2026
- INFORMS Annual Meeting, Atlanta, GA 2025
- ISMS Marketing Science Conference, Washington, D.C. 2025
- Conference on Artificial Intelligence, Machine Learning, and Business Analytics, New Haven, CT 2024

A Causal Approach to Representation Learning for Unstructured Data

- ISMS Marketing Science Conference, Washington, D.C. 2025
- INFORMS Annual Meeting, Seattle, WA 2024
- POMS Annual Conference, Minneapolis, MN 2024
- INFORMS Annual Meeting, Phoenix, AZ 2023

Data-Driven Security Selection for Wealth Management

- POMS Annual Conference, Orlando, FL 2023
- INFORMS Annual Meeting, Indianapolis, IN 2022

Verifying Global Optimality of Candidate Solutions to Polynomial Optimization Problems Using a Determinant Relaxation Hierarchy

- INFORMS Annual Meeting, Anaheim, CA 2021

ACADEMIC
SERVICE

- Reviewer: *Journal of Investment Strategies*
- Session Chair: INFORMS Annual Meeting

REFERENCES

Professor Raphael Thomadsen

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The Winner’s Curse in Data-Driven Decision Making: Evidence and Solutions

Data-driven decision making involves estimating the value of each potential option and selecting the one with the highest estimated efficacy. This approach underpins a wide array of modern marketing, operations, and AI applications, including A/B testing, advertising and bidding, pricing, and personalized targeting. However, several papers have shown that the estimated effectiveness of the chosen options will be systematically overoptimistic (Smith and Winkler 2006, Efron 2011, Andrews et al. 2024), even when the estimated outcomes are themselves unbiased and efficient. Using simulations calibrated to realistic parameter values from recent marketing studies, we first demonstrate that the magnitude of the winner’s curse is often high in relevant marketing contexts, and that the severity of the winner’s curse depends on the true performance difference between options relative to the level of noise in the data, the number of alternatives under consideration, and the number of observations per tested condition. We further show that using machine learning methods to evaluate what treatment to give to individual consumers can lead to extremely high levels of winner’s curse, especially if the machine learning functional form is very flexible. We propose a correction method based on a non-continuous bootstrap, and benchmark our method against several existing proposed solutions across many common marketing scenarios. We demonstrate that our bootstrap approach generally performs well, and usually outperforms the solutions that have been previously proposed in the literature.

SOTA or Luck? The Winner’s Curse in LLM Leaderboards

Public LLM leaderboards are now central to how models are evaluated, compared, and publicized, yet we show that the reported performance of top-ranked models can be systematically overstated. The reason is a leaderboard winner’s curse (WC): benchmark scores are noisy averages over finite task sets, so ranking models in the decreasing order of scores rewards sampling luck alongside genuine ability. This overstatement is large enough to make the apparent winner statistically fragile: across four out of the five SWE-bench variants we study, the published rank 1 model has less than a 50% bootstrap probability of remaining rank 1 under task resampling. We develop an m -out-of- N task bootstrap for iid benchmark tasks while allowing arbitrary within-task correlation among models. We prove that the worst-case rank-induced bias (WC as a special case for the first rank) is $\Theta(1/\sqrt{N})$ in the number of tasks N , and that our correction reduces it to $o(1/\sqrt{N})$. We also introduce a rank probability matrix that replaces a single deterministic ranking with a distribution over plausible rankings. Empirically, rank-1 inflation reaches 1–3 percentage points across eight benchmark settings spanning code generation and agentic customer service. On SWE-bench Verified, selection bias explains 47% of the winner’s apparent top-five lead, and the top 28 models are statistically indistinguishable at the 5% level. Our results suggest that benchmark designers should either enlarge the task pools substantially to suppress the WC or report bias-corrected scores and ranking uncertainty alongside raw rankings.